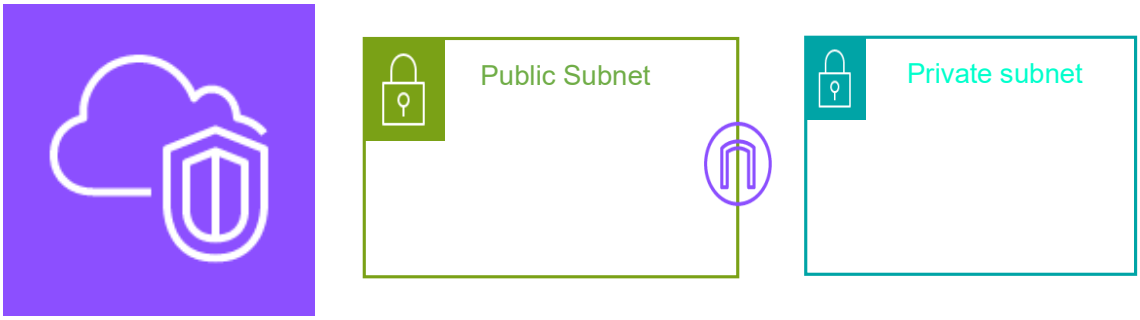
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

Configure VPC using Public & private Subnet in AWS



GOKUL S

Objective

A **VPC (Virtual Private Cloud)** in Amazon Web Services (AWS) is a **logically isolated virtual network** inside AWS where you can launch and manage your cloud resources securely.

- Amazon Virtual Private Cloud (**VPC**) is a service that lets you launch **AWS** resources in a logically isolated virtual network that you define.
- It is logically isolated from other virtual networks in the AWS Cloud.
- A virtual private cloud (VPC) is a secure, isolated private cloud hosted within a public cloud.
- Amazon Virtual Private Cloud (Amazon VPC) hides the complexity, and simplifies the deployment of secure private networks.

Subnets:

A subnet is a sub range of IP addresses in the VPC. AWS resources can be launched into a specified subnet.

- After creating a VPC, you can add one or more subnets in each Availability Zone.



- ✓ *The public subnet* resources that must be connected to the internet.
- ✓ The *private subnet* resources that must remain isolated from the internet.

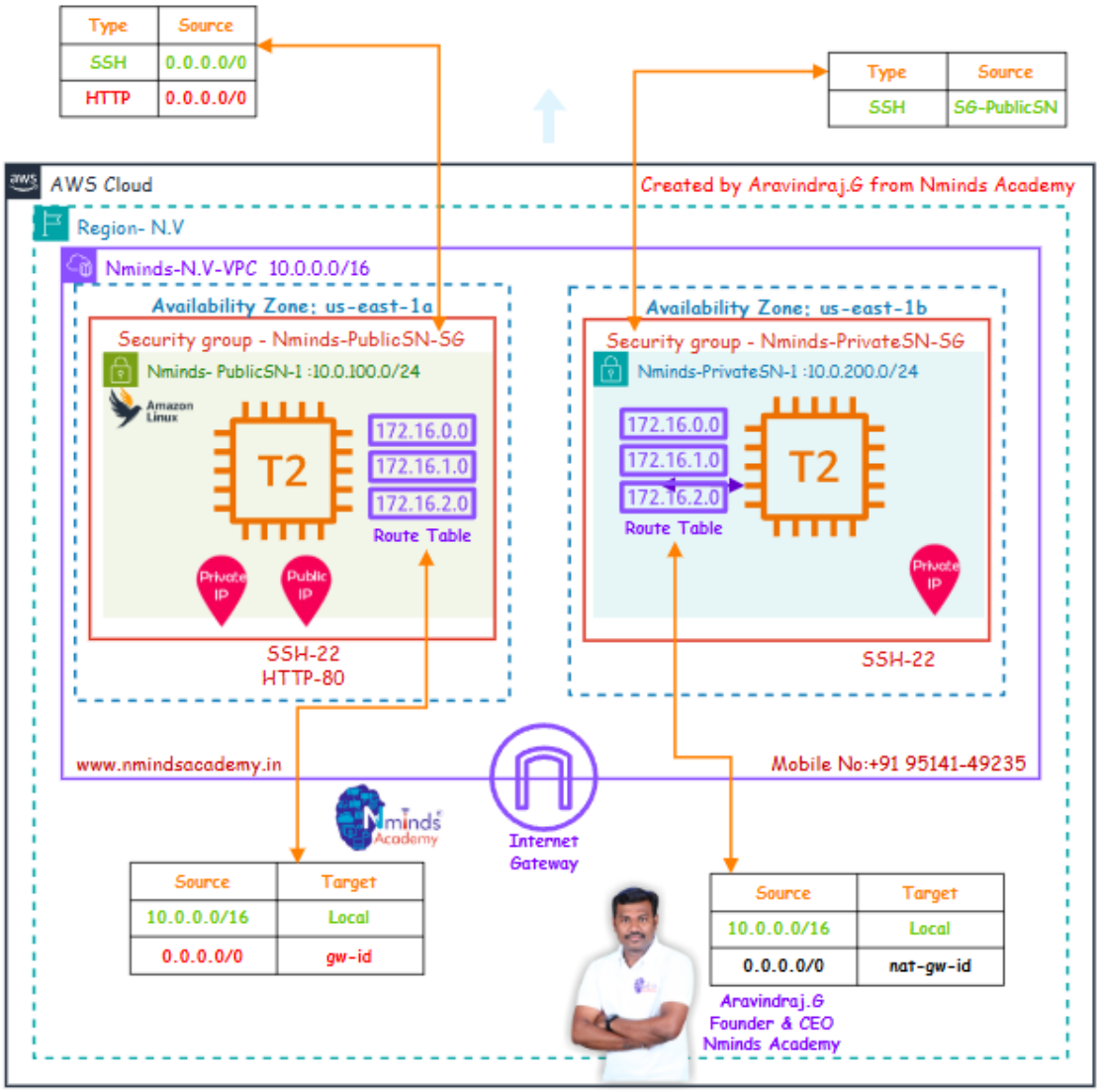
Internet Gateway:

An *internet gateway* is a VPC component that allows communication between resources in your VPC and the internet. It is horizontally scaled, redundant, and highly available. An internet gateway supports IPv4 and IPv6 traffic. An internet gateway serves two purposes. First, it provides a target in your VPC route tables for internet-routable traffic. Second, the internet gateway performs network address translation (NAT) for instances that were assigned public IPv4 addresses.



Topology

How to create custom VPC with Public Subnet and Private Subnet in AWS



Used Services in AWS

Amazon VPC

Amazon Elastic Compute Cloud (Amazon EC2)

Used Services in Windows Server

APACHE HTTP SERVER

Apache Webserver

Public subnet

www.nminds academy.in

Internet gateway

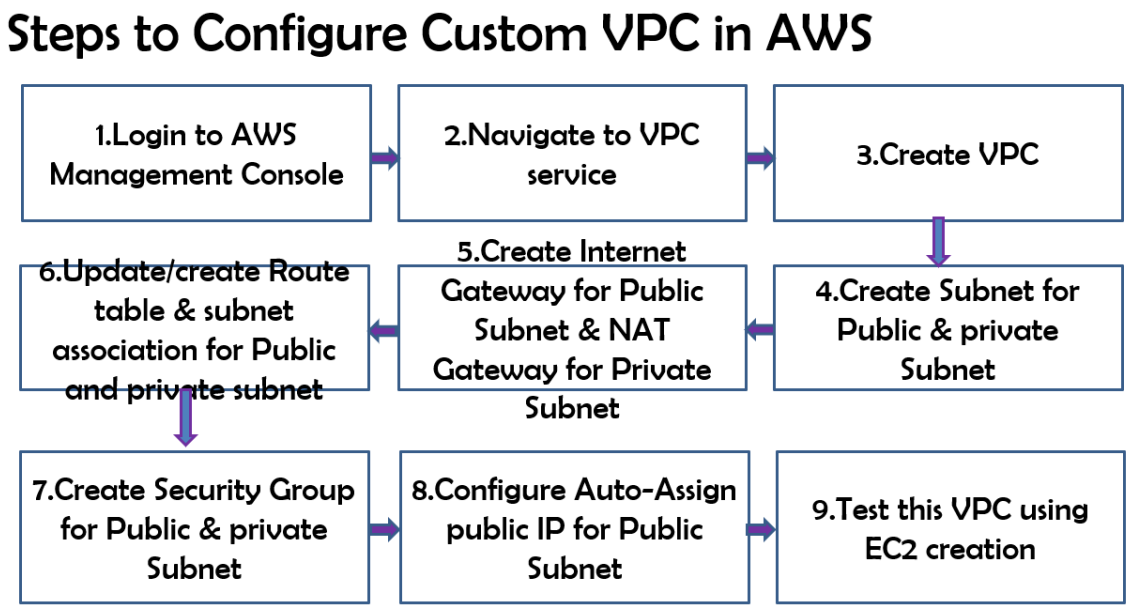
172.16.0.0
172.16.1.0
172.16.2.0

Route table

+91-95141 49235

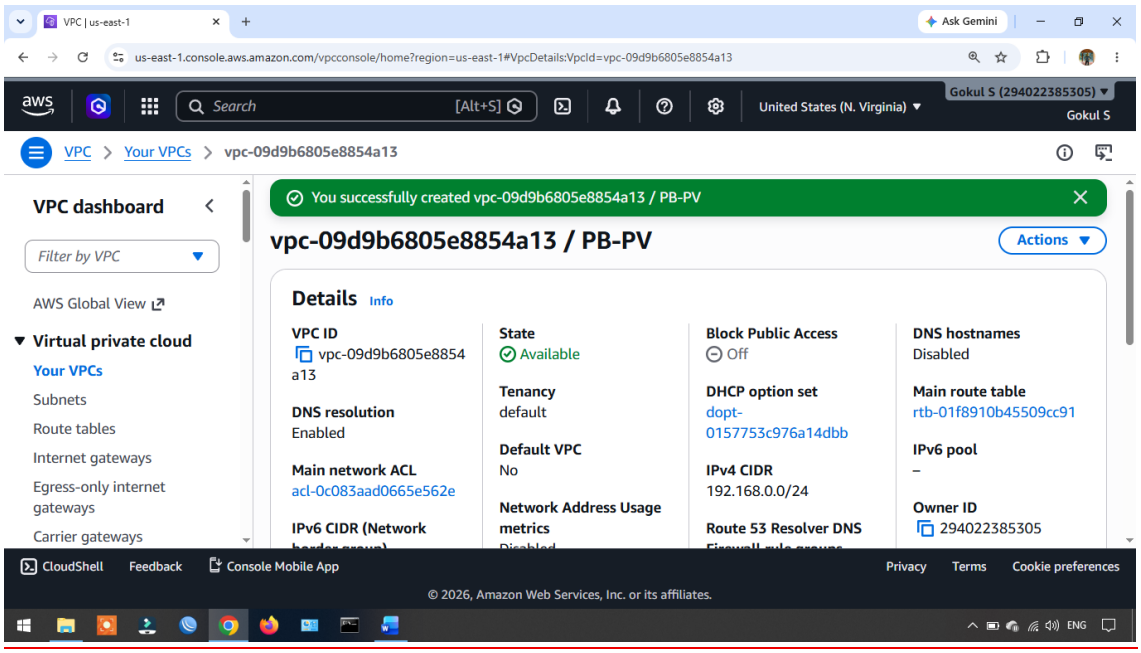
NmindsAcademy

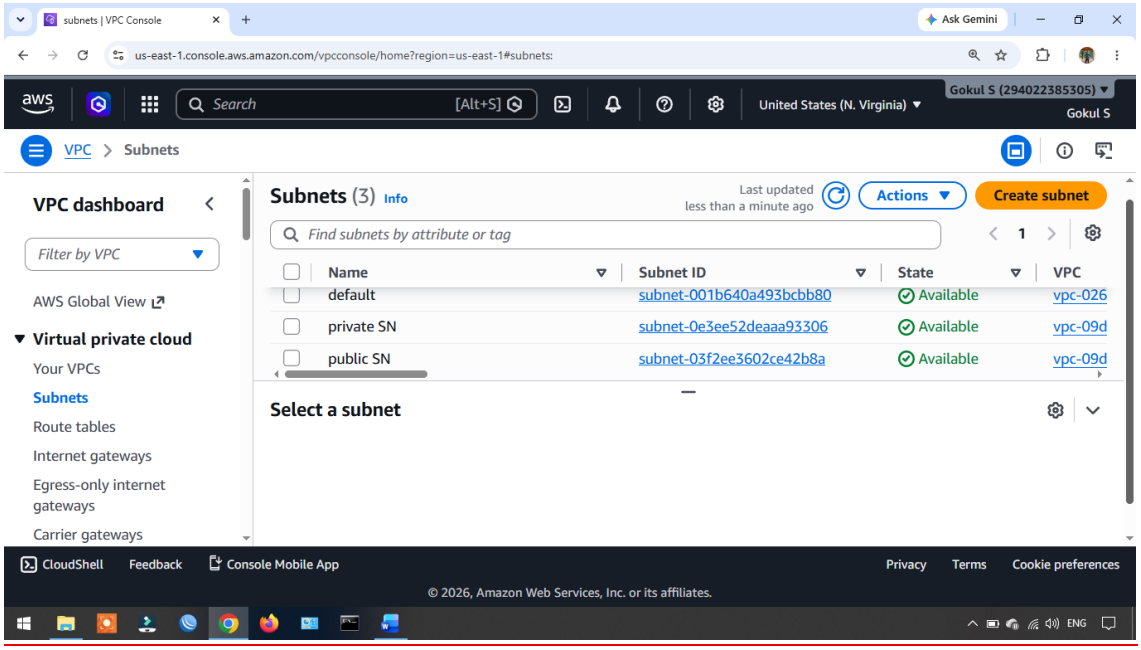
Execution Steps:

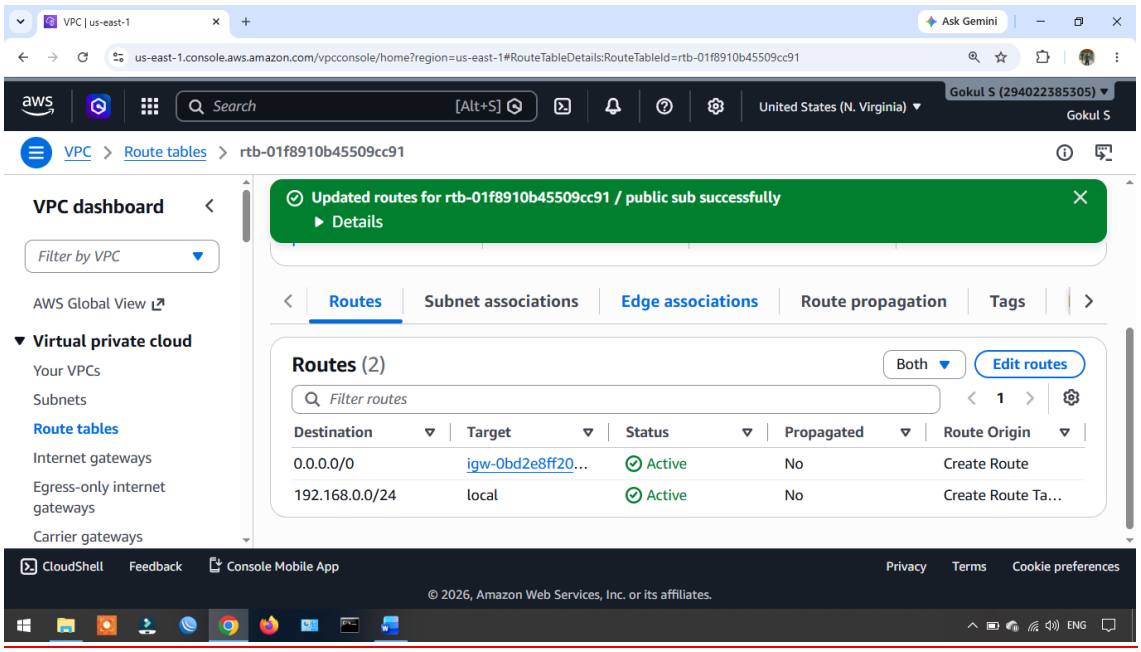


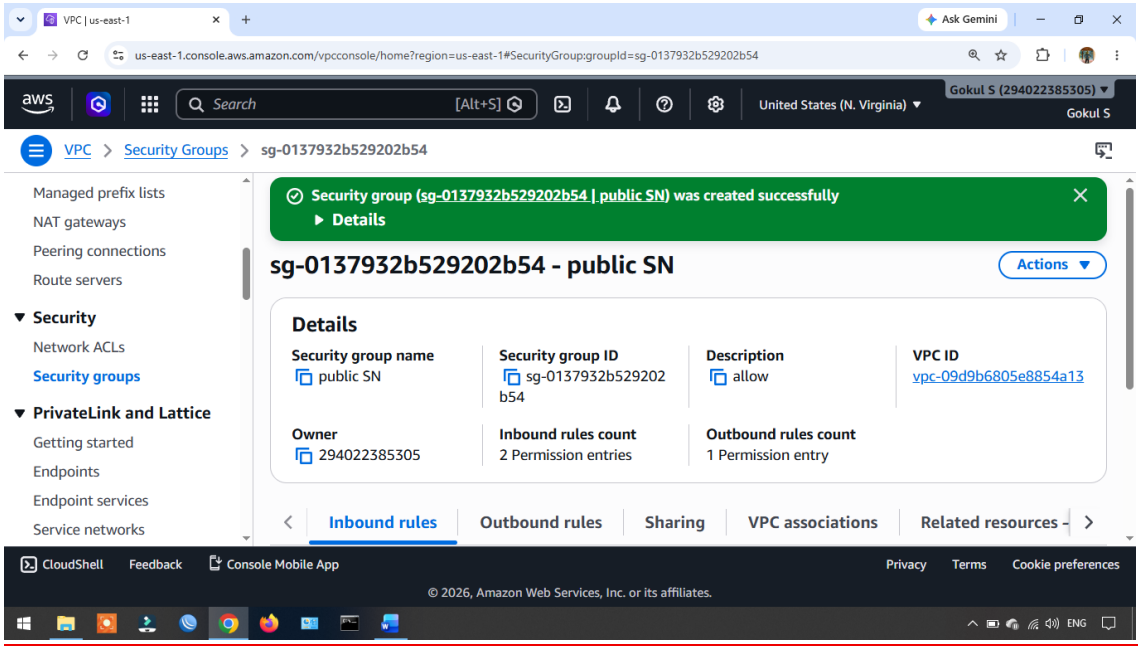
Execution Tasks:

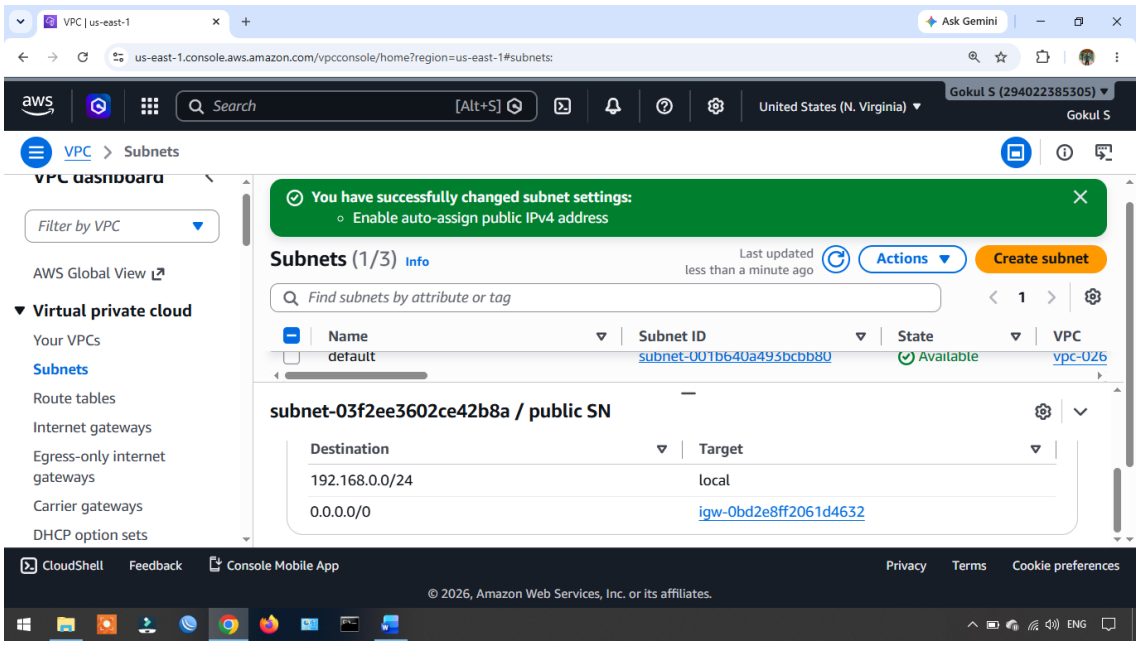
Task1: Create the custom VPC with Public Subnet & private subnet based on the above topology:

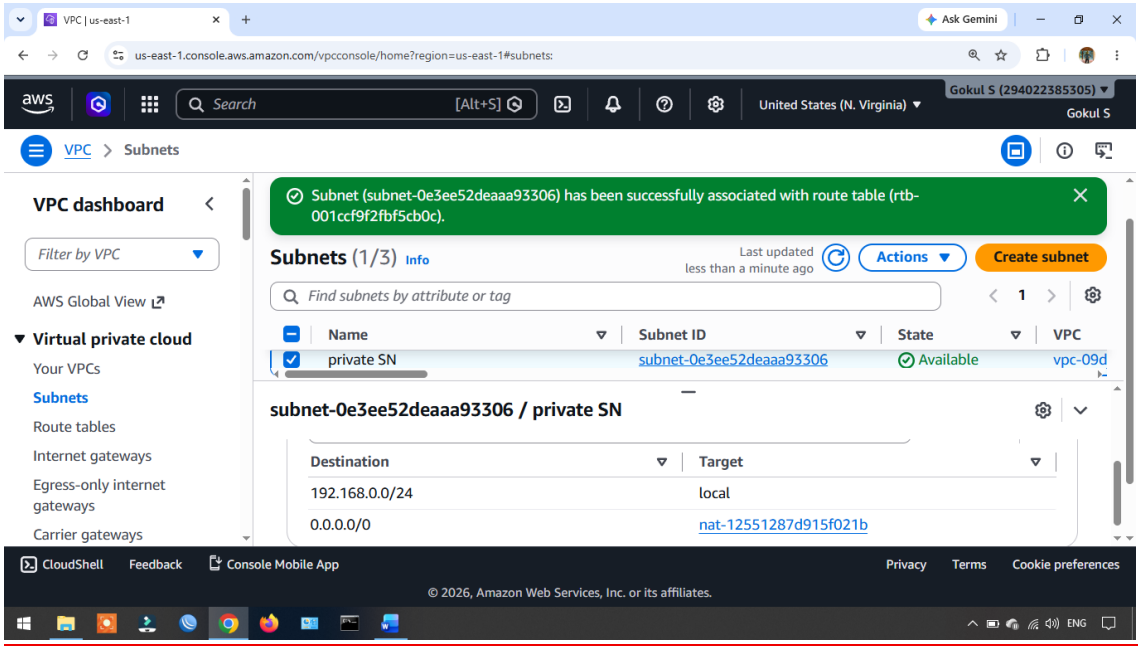



subnets | VPC Console
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Subnets
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
Subnets (3) Info
Last updated less than a minute ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
default
subnet-001b640a493bcbb80
Available
vpc-026
private SN
subnet-0e3ee52deaaa93306
Available
vpc-09d
public SN
subnet-03f2ee3602ce42b8a
Available
vpc-09d
Select a subnet
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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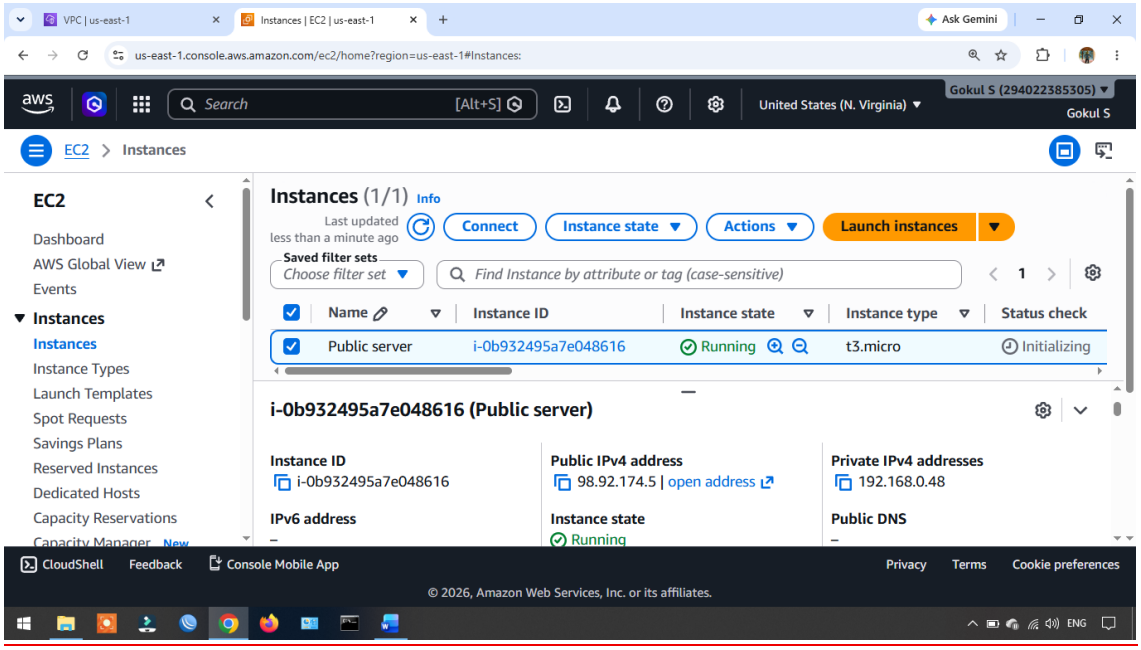

VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTableDetails:RouteTableId=rtb-01f8910b45509cc91
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Route tables > rtb-01f8910b45509cc91
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
Updated routes for rtb-01f8910b45509cc91 / public sub successfully
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
igw-0bd2e8ff20...
Active
No
Create Route
192.168.0.0/24
local
Active
No
Create Route Ta...
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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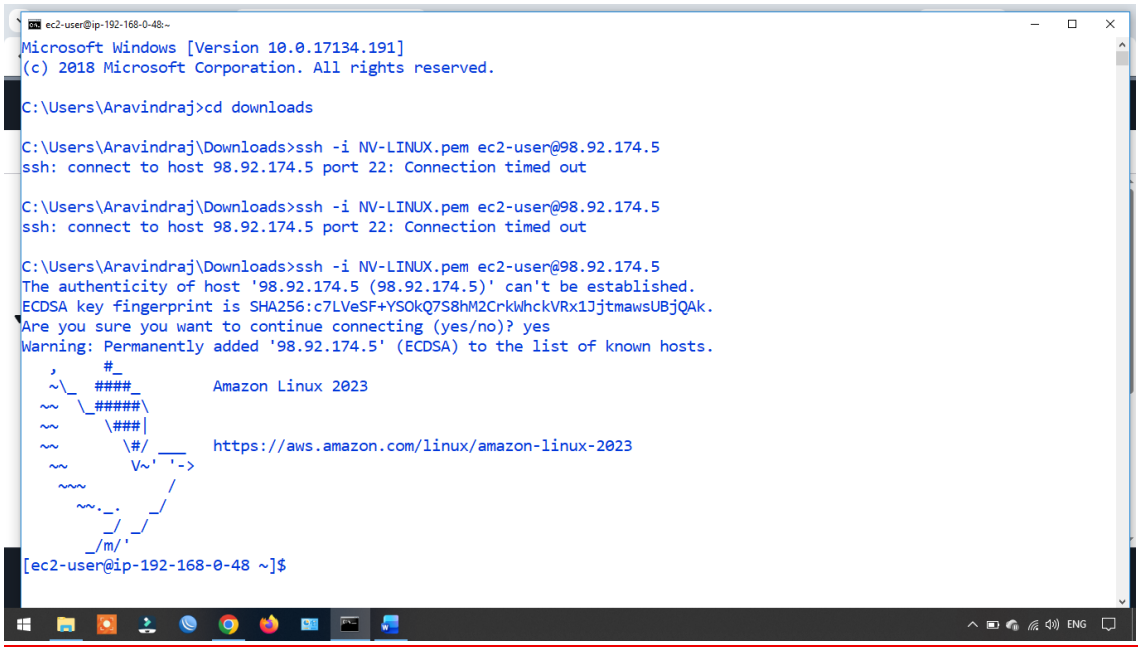

VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#SecurityGroup:groupId=sg-0137932b529202b54
Search [Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Security Groups > sg-0137932b529202b54
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Security group (sg-0137932b529202b54 | public SN) was created successfully
Details
sg-0137932b529202b54 - public SN
Actions
Details
Security group name
public SN
Security group ID
sg-0137932b529202b54
Description
allow
VPC ID
vpc-09d9b6805e8854a13
Owner
294022385305
Inbound rules count
2 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources
CloudShell
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:
Search [Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Subnets
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
You have successfully changed subnet settings:
Enable auto-assign public IPv4 address
Subnets (1/3) Info
Last updated less than a minute ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
default
subnet-001b640a493bcbb80
Available
vpc-026
subnet-03f2ee3602ce42b8a / public SN
Destination
Target
192.168.0.0/24
local
0.0.0.0/0
igw-0bd2e8ff2061d4632
CloudShell
Feedback
Console Mobile App
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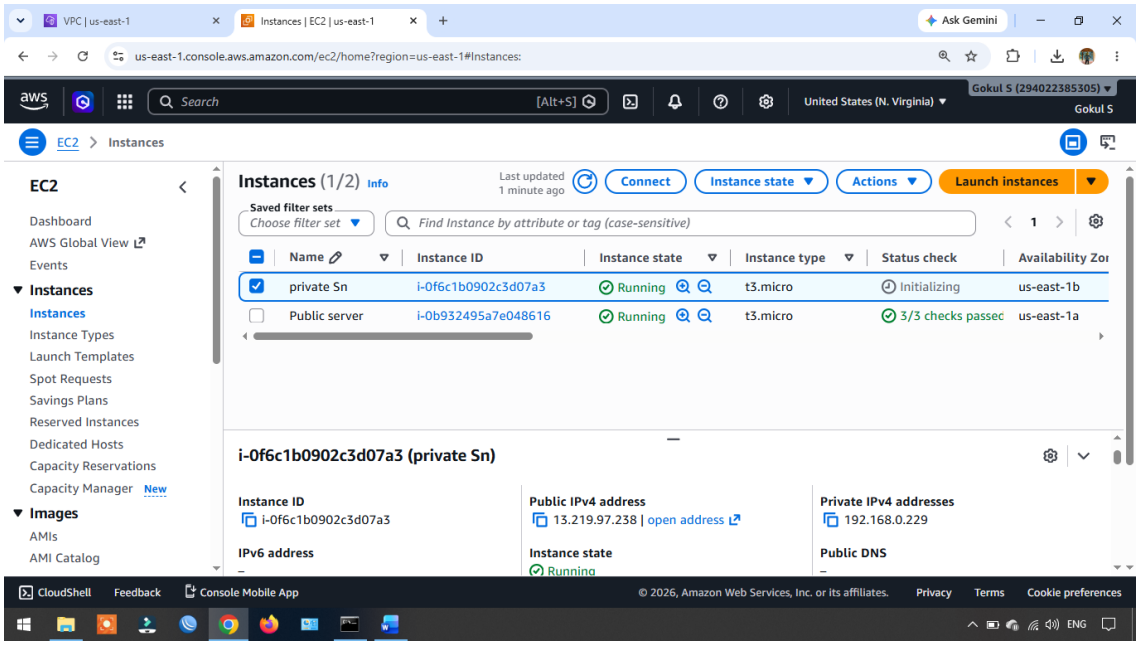


Task2: Create Linux Instance & Install Apache Webserver (EC2) on newly created public subnet:





Task3: Create Linux Instance (EC2) on newly created private subnet:



Task4: The Private subnet Linux machine only able to access by public subnet instance only, how to configure for this scenario with execution steps

